

Fig. 1 MICrONS: from one imaged cortical volume to a label-free per-neuron decoding task

a One mouse, one mm³ of visual cortex

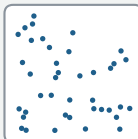
MICrONS release (Ding et al. 2025)

13 two-photon scans; V1, RL, AL, LM

12,894 neurons co-registered

to the EM volume

each with responses, orientation,
receptive field, soma position



b Relations: correlations between neurons

each neuron = its response vector to a stimulus all neurons saw:

in vivo: trial-averaged responses to the oracle movie clips (120 bins)

digital twin: model responses to a shared movie (4,999 bins → 512 PCs)

Gram entry (i, j) = correlation of neuron i's and neuron j's vectors

12,894 × 12,894 matrix; stimulus identity never enters the decoder

c Contents (training targets only)

preferred orientation: in vivo, 8 classes,

gOSI ≥ 0.25 → 5,287 labelled neurons

receptive-field centre (x, y): twin STA fit,

test correlation ≥ 0.2 → 11,326 labelled

unlabelled neurons stay in as tokens

d Protocol: halve the neurons, sample populations, decode each neuron from its row of the population's Gram

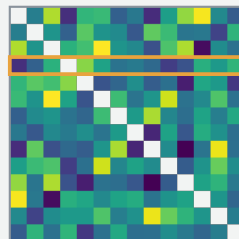
all 12,894 neurons, random half split



populations of 512 are sampled inside a half,
never across the two halves

sample
512

population Gram (512 × 512)



one row = one neuron's token:
its correlations to the other 511

transformer

rows as tokens

attention over the population

2M-57M parameters

per-neuron prediction

orientation class or (x, y)

loss on labelled training neurons

scored on test neurons only

epoch chosen on a held-out slice

of the training half, never on test

Same brain, single-animal populations, new neurons: the `within` cell of the Allen regime table (Fig. 4).